

Press Release (Future Customer Announcement)

For Immediate Release

MyVeloFit Introduces Advanced Posture Detection for Smarter Bike Fit Analysis

[DATE] – MyVeloFit, the leader in AI-driven bike fitting, is taking bike fit analysis to the next level with the launch of its **Posture Detection** feature. This new tool enhances traditional joint-angle-based assessments by analyzing **rider posture**, providing deeper insights into comfort, efficiency, and injury prevention.

Cycling performance isn't just about angles—it's about how you hold yourself on the bike. With **Posture Detection**, MyVeloFit now evaluates **sacral angle** and **spinal posture**, helping riders and fitters better understand the impact of their position. By leveraging advanced image processing and AI, this feature identifies key posture markers in a still frame, delivering **precise, actionable fit recommendations**.

Key Benefits:

- ✓ **Sacral Angle Measurement** – Analyze pelvic position to assess core stability and lower back engagement.
- ✓ **Spinal Posture Segmentation** – Detect upper and mid-back positioning to optimize aerodynamics and long-ride comfort.
- ✓ **AI-Driven Precision** – Uses advanced segmentation and background removal to isolate the rider from distractions.
- ✓ **Actionable Fit Insights** – Provides measurable, easy-to-understand posture metrics for fine-tuning your fit.

"Posture is critical to cycling efficiency and comfort, yet it's often overlooked in traditional bike fitting. With this new feature, we're helping cyclists of all levels achieve a more sustainable and powerful riding position," said [NAME], CEO of MyVeloFit.

Posture Detection is currently available for internal analysis and will be released to users in future updates. Stay tuned for more ways MyVeloFit is redefining the future of bike fit.

About MyVeloFit

MyVeloFit is the world's most advanced AI-driven online bike fitting platform, helping cyclists optimize their riding position, improve comfort, and prevent injuries—all from the convenience of their home.

For more information, visit www.myvelofit.com

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Customer FAQs (External)

Q: What is MyVeloFit's new Posture Detection feature?

A: Posture Detection enhances MyVeloFit's AI bike fit by analyzing **sacral angle and spinal posture**, providing deeper insights into your riding position beyond traditional joint angles.

Q: Why does posture matter in bike fit?

A: Your posture affects power transfer, aerodynamics, and long-term comfort. Poor spinal alignment can cause **lower back pain, neck strain, and fatigue**—issues that Posture Detection helps identify and correct.

Q: How does Posture Detection work?

A: Using **AI-driven image processing**, MyVeloFit analyzes a single still frame from your bike fit video. It isolates your **sacral angle** and **spinal posture** by segmenting your back into multiple sections and measuring the angles.

Q: Will this feature be available in my fit results?

A: Not yet! For now, Posture Detection is being tested internally and will be added to user profiles in a future update.

Q: Can this feature analyze my past bike fits?

A: Yes! We are working on batch processing to allow **posture analysis on previous fits**, so you can track changes over time.

Q: What's next for this feature?

A: Future updates may include **knee tracking wave and pedaling speed analysis**, helping you fine-tune your fit even further.

Internal FAQ (For MyVeloFit Team Use)

Q: How does the system detect sacral angle?

A: The AI isolates the rider using background removal and segmentation. It then:

1. Identifies the **hip marker** and traces horizontally until it detects the background.
2. Takes a point $\frac{1}{4}$ of the way between the **hip and shoulder markers** and applies the same logic perpendicularly.
3. Measures the **sacral angle** based on these points.

Q: How does it measure spinal posture?

A: The AI tracks the rider's **back shape** from the hip to the shoulder, then splits the spine into three segments:

1. **Lower Back (Sacral Angle to Mid-Spine)**
 2. **Mid Back (Mid-Spine to Upper Back)**
 3. **Upper Back (Upper Back to Shoulder Marker)**
- Each segment is analyzed for curvature and optimal alignment.

Q: How will this data be presented?

A: The **measured angles** will be outputted in JSON format and visualized with overlay lines on the admin dashboard. This feature will not be visible to users initially.

Q: Will cluttered backgrounds affect accuracy?

A: Possibly. While AI segmentation is designed to isolate the rider, excessive background noise may impact detection. Fine-tuning may be needed.

Q: How will we validate accuracy?

A: We will **batch process past fits** (e.g., 1,000+ samples) to analyze trends, refine detection methods, and compare results with expert manual fits.

Q: What are the next potential enhancements?

A: Future improvements could include:

- **Live posture tracking** rather than single-frame analysis.
- **Advanced analysis tools** like knee tracking wave and pedaling efficiency metrics.